Week 4 Lecture

Topic: Reading and writing about movement

Part 1: Grammar – Prepositions of Movement

Language Reference

Preposition	Meaning	Engineering example
from … to	origin → destination	Heat is transferred **from** the engine **to** the coolant.
into	entering a closed space or substance	Fuel is injected **into** the cylinder.
out of	leaving a closed space	Steam escapes **out of** the boiler.
onto	moving to a surface	The robot places the chip **onto** the circuit board.
off	leaving a surface	The part slides **off** the conveyor belt.
through	passing inside something (pipe, tunnel, filter)	Current flows **through** the resistor.
across	from one side of a surface or line to the other	The cable runs **across** the ceiling.
over	above and across an obstacle	The crane lifts the beam **over** the wall.
under	below and across	The pipeline passes **under** the river.
along	following the length of something	Sensors are placed **along** the pipeline.
around	in a circle, or avoiding an obstacle	The robot moves **around** the machine.
past	going by without stopping	The train goes **past** the station.
up / down	higher or lower along a path	The piston moves **up** and **down** inside the cylinder.
toward(s)	in the direction of (may not arrive)	The antenna turns **toward** the satellite.

Tricky Pairs:

- **Through vs across.** *Through* is 3D (inside something with walls: a pipe, a filter, a tunnel). *Across* is 2D (a surface or line). Memory tip: you go *through* a tunnel, *across* a bridge.
- **Over vs across.** Both can describe crossing a river. *Over* stresses being above; *across* stresses reaching the other side.
- **In vs into.** *In* shows position (*The liquid is in the tank.*). *Into* shows movement (*The liquid flows into the tank.*).
- **Out of vs from.** *Out of* is for leaving an enclosed space; *from* is for a starting point or source.

Common Errors:

Incorrect	Correct
The gas **enters to** the tank.	The gas **enters** the tank. / The gas **goes into** the tank.
The engineer went **in** the lab.	The engineer went **into** the lab.
The wire goes **across** the hole in the wall.	The wire goes **through** the hole in the wall.
Water flows **out from** the pipe.	Water flows **out of** the pipe.

Remember: the verb **enter** never takes a preposition. *The gas enters the tank.*

Practice Activities:

Activity 1 – Gap fill

Complete each sentence with a word from the box. Two words are extra.

> **into · out of · through · along · across · up · down · around · past · onto · over · toward**

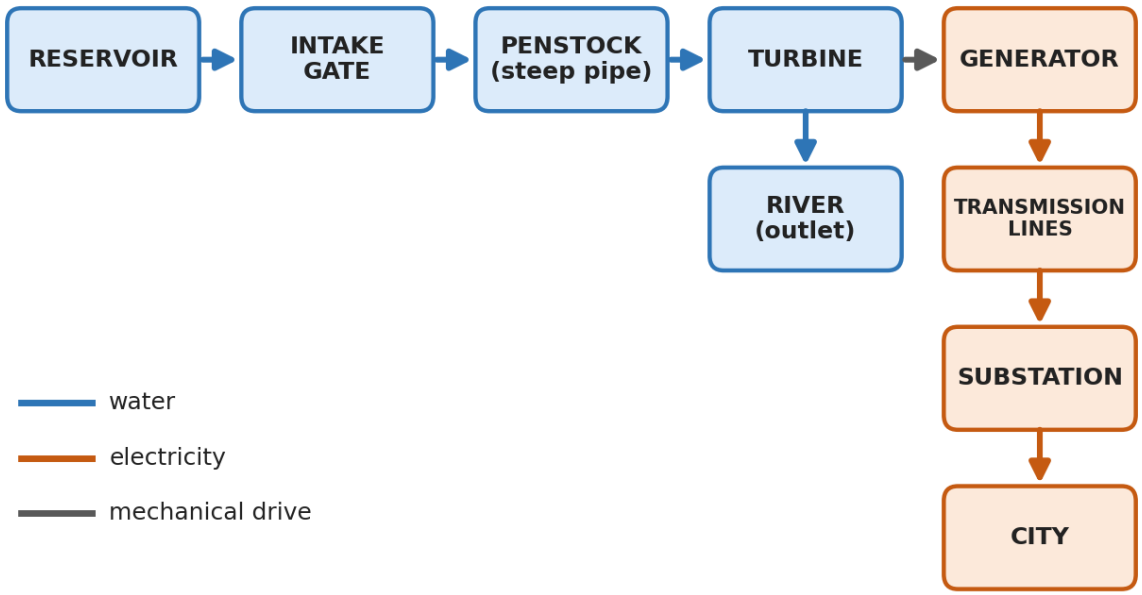
1. Hot gases leave the engine _____ the exhaust pipe.
2. The technician lifted the motor _____ the workbench.
3. Cables run _____ the wall inside a plastic conduit.
4. The crane swings the load _____ the fence.
5. Pour the liquid slowly _____ the beaker.
6. When pressure is too high, the valve lets steam _____ the boiler.
7. The robot goes _____ the obstacle instead of hitting it.
8. The elevator moves _____ to the top floor.
9. The freight train rushed _____ the platform without stopping.
10. The drone flew _____ the river to the other bank.

Activity 2 – Choose the correct option

Circle the correct preposition.

1. The engineer walked **(in / into)** the control room and switched on the monitors.
2. Light travels **(across / through)** the optical fibre.
3. The ship sailed **(through / across)** the Atlantic Ocean.
4. The finished part slides **(off / out of)** the end of the conveyor belt.
5. A truck drove **(under / below)** the bridge.
6. Data moves **(from / of)** the sensor **(to / at)** the server.

Activity 3 – Describe the diagram



Speaking. Partner A describes how the water moves through a hydroelectric plant. Partner B follows the diagram and corrects mistakes. Then swap: Partner B describes the path of the electricity.

Writing. Write **six sentences** about the plant. Use at least **six different** prepositions of movement.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

Activity 4 – Program the robot

Student A: turn to the **Partner A card**. **Student B:** turn to the **Partner B card** (both at the back of this worksheet).

- A reads the robot's route aloud, one step at a time. B listens and draws the path on the map. **Do not look at each other's cards.**
- When A has finished, compare B's drawing with A's instructions. Did the robot end in the right place?
- **Round 2:** swap roles. Each student writes their own 5-step route (in the classroom or on campus) and dictates it to their partner.

Activity 5 – Error correction

Find and fix the mistake in each sentence. Write the correct sentence on the line.

1. The gas enters to the tank through a valve.

   _____

2. The cable passes across a small hole in the wall.

   _____

3. The engineer went in the lab to check the equipment.

   _____

4. The water goes out from the pipe at high pressure.

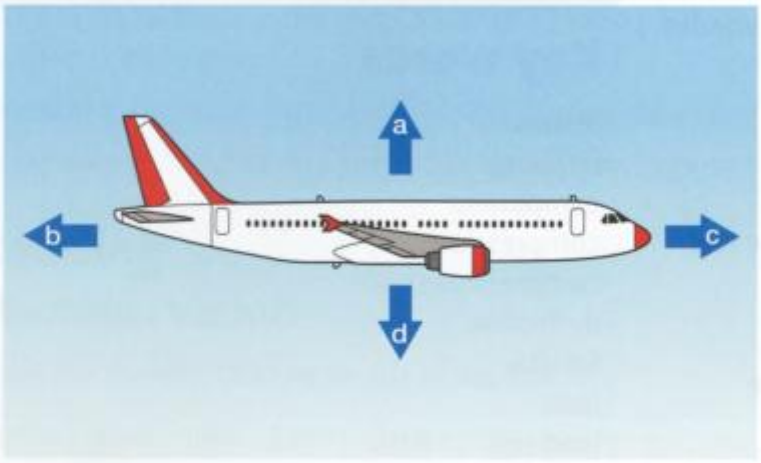
   _____

5. The technician fell down from the ladder.

   _____

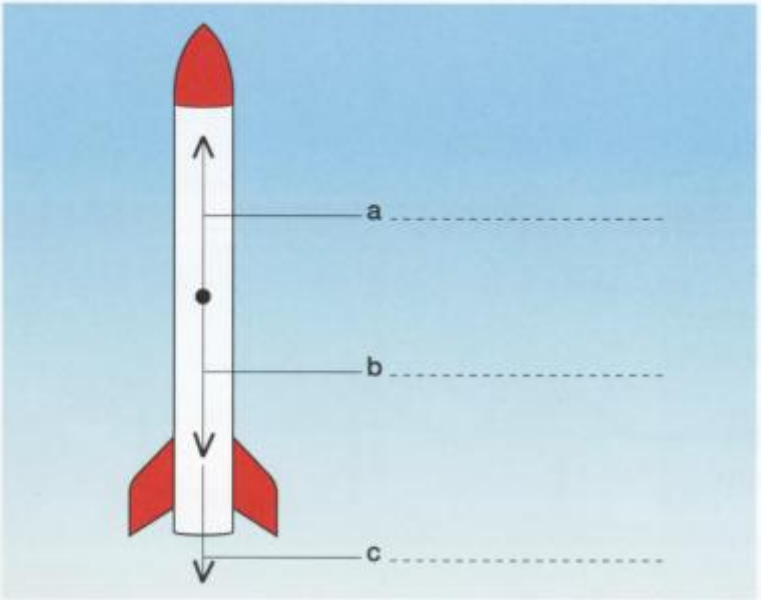
Part 2: Reading and Listening Comprehension – Movement

1 Match the words with the arrows on the diagram.



thrust	the force produced by an engine that pushes a vehicle forward
drag	the force of the air that acts against the movement of an aircraft
lift	the upward pressure of air on an aircraft when it is flying
weight	the downward pull of gravity

2 Label the three forces on the rocket.



3 Work in pairs. Answer the questions.

1 Which force acts on the plane but not on the rocket?
2 Can the plane fly if the engine stops supplying thrust?
3 Can the rocket fly if the engine stops supplying thrust?

Vocabulary: Forces and Motion

1 Look at the definitions. Match each definition with an example below.

speed	how fast something is moving, measured as the distance travelled in a unit of time
velocity	speed of travel in a given direction
acceleration	the increase of velocity in a unit of time
deceleration	negative acceleration (or the decrease of velocity in a unit of time)
equilibrium	a condition where forces are balanced

1 The train is going 120 kilometres per hour straight ahead.
2 The brakes can stop the car in one second when it is travelling at 45 kilometres per hour.
3 The motorcycle is travelling at 20 kilometres per hour.
4 The car isn't moving. It's sitting still.
5 The car goes from 0 to 100 kilometres per hour in 3.5 seconds.

2 Complete the sentences with words from **1**.

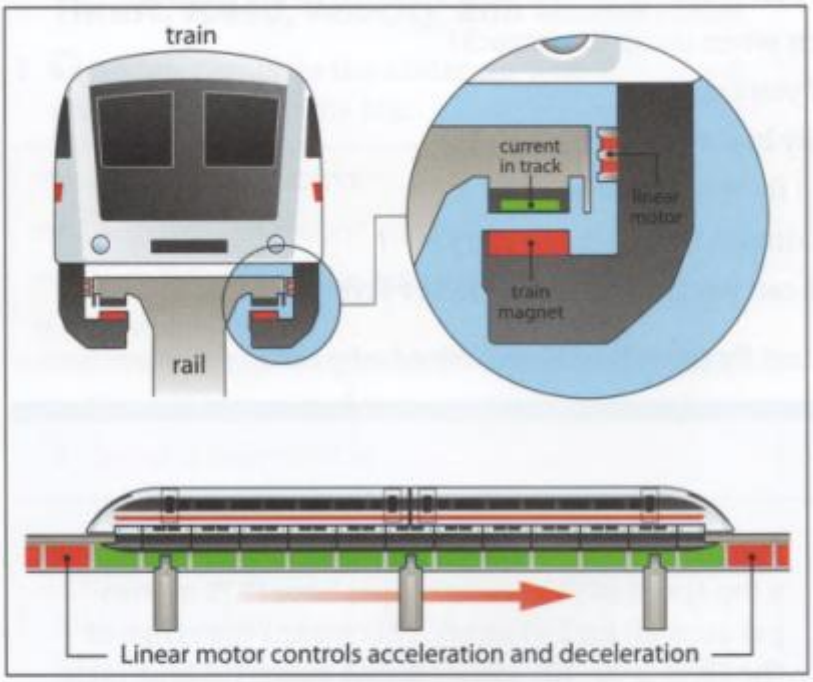
1 When a rocket takes off, _____ occurs.
2 When a car hits a wall, there is a very rapid _____.
3 A building is a good example of _____.
4 One of the most important instruments in a car tells you the car's _____.
5 If you don't know which direction something is travelling, then you can't know its _____.

Listening: Maglev Trains

1 Look at the picture. What do you know about maglev technology?



2 Look at the diagram. Complete the description with the words in the list.



magnets motor poles track train wheels

The train doesn't have _____[1]. Pairs of magnets with the same _____[2] keep the train above the _____[3]. It also has no engine. A linear _____[4] in the track pulls guide _____[5] on the train and makes it move. The system also stops the _____[6].

3 Listen. Write T (true) or F (false).

1 Stefan Urbach has been working on a rail system at an airport.
2 The project is on budget but somewhat behind schedule.
3 The maglev system at Birmingham Airport was very successful.
4 Maglev trains don't need a lot of maintenance.
5 For this project, speed is more important than acceleration.
6 The technology is called EMS – electromagnetic suspension.
7 The linear motor is made up of coils of wire under the track.
8 The control system is extremely simple.
9 The system was expensive to install but it will be economical to run.
10 Deceleration uses a lot of energy.

4 List the benefits and drawbacks of the maglev system. Listen again and check your answers.

5 Do you know of any other maglev train projects?

Reading

The JetLev-Flyer

1 Look at the picture. What can you see? How do you think this device works?

2 Read the text and check your answers.

3 Answer the questions.

1 Why aren't the pump, motor, or fuel tank attached to the jetpack?
2 How do you turn when using the jetpack?
3 Why is it safe if you crash?
4 Why can't you fly higher than ten metres?
5 How far can you fly with the jetpack?
6 What is the maximum weight it can carry?
7 What other uses can you think of for the JetLev-Flyer?

4 Would you like to test fly the JetLev-Flyer? Why / why not?

The JetLev-Flyer



The JetLev-Flyer can fly ten metres above water at a top speed of 35 kilometres per hour (9.75 metres per second) for 2–3 hours. This means the range of the Flyer is 70–105 kilometres. A powerful four-cycle engine produces 1900 newtons of thrust for stable, controlled flight.

The flight controls

All of the controls are onboard the jetpack. You control thrust by twisting the hand grip, as on a motorcycle. To move forward and accelerate, you push your arms down. To decelerate or reverse, you pull up. Or you can keep the water jet pointed straight down and hover, floating above the water in equilibrium. You can turn either by angling the water jets in different directions or by shifting your weight from side to side.

Safety features

You can fly it only over water, so that makes it safe to begin with – the water underneath you will usually allow a soft landing. It also has a five-point quick-release harness and good head and back support. Finally, the pack floats.

How it works

The jet propulsion medium is water, which means the jetpack uses a high-powered stream of water to create thrust. Water is supplied at high flow but low pressure to the jetpack by way of a ten-metre hose, which is attached to a boat that carries the motor and pump.

It will carry a pilot weighing up to 150 kilograms.

At the moment it is being marketed for water sports.

Number talk

Thrust, speed, velocity, and acceleration

1 Listen. Complete the sentences with the correct abbreviation from the box.

m/s = metres per second
m/s^2 = metres per second squared
m/s (+ direction) = metres per second (+ direction)
N = newtons

1 Thrust is expressed in _____.
2 Speed is expressed in _____.
3 Velocity is expressed in _____.
4 Acceleration is expressed in _____.

2 Match the questions with the answers.

1 How much thrust does this jet engine produce?	a 70 m/s
2 How fast is the train going?	b 90 m/s to the north
3 What's the velocity of the plane?	c 8700 N
4 What's the acceleration of the car?	d 3 m/s^2

3 Listen and check your answers.

Part 3: Written Expression – Explaining Technical Specifications

Look at the picture and the technical information. Using the introduction of the JetLev-Flyer text on p.30 as a model, write a description.



Martin Jetpack

Maximum height for flying	2400 mm above the ground
Top speed	100 km/h
Range	50 km
Thrust	2670 newtons
Engine	Two-cycle – Martin Aircraft 2.0 litre
Safety features	harness, parachute
Empty weight	115 kg
Maximum pilot weight	130 kg